



PHYSICS SSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION – A (Marks 12)

Time allowed: 15 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Version No.				
1	0	0	4	1

①	●	●	①	①
●	①	①	①	●
②	②	②	②	②
③	③	③	③	③
④	④	④	●	④
⑤	⑤	⑤	⑤	⑤
⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨

ROLL NUMBER						

①	①	①	①	①	①	①
①	①	①	①	①	①	①
②	②	②	②	②	②	②
③	③	③	③	③	③	③
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⑥	⑥	⑥	⑥	⑥	⑥	⑥
⑦	⑦	⑦	⑦	⑦	⑦	⑦
⑧	⑧	⑧	⑧	⑧	⑧	⑧
⑨	⑨	⑨	⑨	⑨	⑨	⑨

Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. A device measures a charge of 3000 nC. Convert it into microcoulombs (μC).	0.03 μC	3 μC	30 μC	300 μC	☐	☐	☐	☐
2. A cyclist moves 2 km in 4 minutes. What is the speed in m/s?	4 m/s	8.3 m/s	12 m/s	16.3 m/s	☐	☐	☐	☐
3. Which fundamental force is the strongest of all?	Gravitational	Electro-weak	Strong force	Magnetic	☐	☐	☐	☐
4. What is a reasonable estimate for the mass of a physics textbook?	2 mg	1 kg	10 kg	1 g	☐	☐	☐	☐
5. Which unit belongs to a derived quantity?	Second (s)	Kelvin (K)	Meter (m)	Newton (N)	☐	☐	☐	☐
6. If two unequal forces act parallel but in opposite directions, the forces are:	Unlike parallel forces	Balanced forces	Like parallel forces	Centripetal forces	☐	☐	☐	☐
7. A hypothesis is best described as:	A testable prediction	A universally accepted fact	A mathematical equation	A proven theory	☐	☐	☐	☐
8. When a soft iron piece is placed near a magnet, it:	Becomes a permanent magnet	Loses its mass	Gets temporary poles (N and S)	Repels the magnet	☐	☐	☐	☐
9. Why does a heavy object resist change in its motion?	It has greater inertia	It has more speed	It has more friction	It has no force	☐	☐	☐	☐
10. If a man applies 50 N of force to lift a load 0.5 m upward then what is the work done?	10 J	20 J	25 J	50 J	☐	☐	☐	☐
11. Which one of the following is the correct formula for magnetic field of a solenoid?	$B = \frac{\mu n}{\ell}$	$B = \mu_o n \ell$	$B = \frac{n \ell}{\mu_o}$	$B = n_o \ell$	☐	☐	☐	☐
12. A scientist who studies matter, energy, and their interactions is called a:	Chemist	Biologist	Engineer	Physicist	☐	☐	☐	☐

—1SA-I 26004-(B)—

• $v = \frac{d}{t}$ • $W = \vec{F} \cdot \vec{d}$ • $w = mg$

**PHYSICS SSC-I**

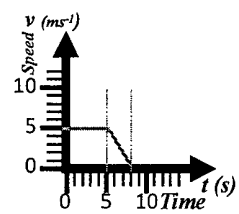
Inclusive Scheme of Studies 2024 (1200 Marks)

Time allowed: 2:45 Hours

Total Marks Sections B and C: 48

SECTION – B (Marks 30)**Q. 2 Attempt the following parts.****(10 x 3 =30)**

(Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)

(i)	What is meant by density? Also write its formula and unit.	1x3	OR	Briefly describe scalar and vector quantities with one example each	1.5 +1.5
(ii)	Why soft magnetic materials, such as soft iron, are used to provide shielding from magnetic fields? Explain briefly.	03	OR	Sketch distance–time graph of an object when it is: a. At rest, b. Moving with constant speed, and c. Accelerating	1+1 +1
(iii)	What is average orbital speed? Why increasing the orbital radius increases the total distance traveled in one orbit?	1+2	OR	Briefly describe the following branches of physics: a. Optics b. Cosmology c. Biophysics	1x3
(iv)	A bicycle rider leans downward and widens the stance while turning. How these actions help prevent the bicycle from falling? Explain briefly.	03	OR	From given speed time graph, calculate: a. Total distance covered b. Acceleration in first 5 seconds	1.5 + 1.5
					
(v)	State the relationship between force and extension for a spring according to Hooke's Law.	03	OR	Differentiate between center of mass and center of gravity.	1.5 + 1.5
(vi)	How changes in force and mass can affect the acceleration of an object? Explain briefly in light of Newton's second law.	1.5+ 1.5	OR	A cyclist applies brakes and slows down from 12ms ⁻¹ to 4ms ⁻¹ in 3 seconds. Calculate the acceleration.	03
(vii)	What is plasma? How it differs from ordinary gases?	2+1	OR	Compare temporary magnets and permanent magnets. Give one example of each.	1.5 +1.5
(viii)	Under what conditions Newton's Laws do not remain applicable? (Give two conditions)	1.5 +1.5	OR	How the Earth's magnetic field protects life on the planet from harmful cosmic radiation? Explain briefly.	03
(ix)	A digital balance gives slightly different readings each time a stone is placed on it: a. Identify the error. b. How can this error be fixed?	1+2	OR	A 2 kg ball changes its velocity from 4ms ⁻¹ to 10ms ⁻¹ . Calculate the impulse applied to the ball.	03
(x)	Prove that $K.E = \frac{1}{2}mv^2$	03	OR	Compare precision and accuracy in measurements.	03

SECTION – C (Marks 18)**Note: Attempt the following questions.****(3x6=18)**

Q.3	Describe the three states of equilibrium. For each give one clear daily-life example.	2x3	OR	On what principle hydraulic brakes in vehicles work? How the system multiplies force to stop the vehicle effectively? Make a neat sketch.	1+4 +1
Q.4	A roller coaster car of mass 500 kg starts from rest at the top of a hill 20 m high. a. Calculate its Gravitational PE at the top. b. Assuming no friction, calculate its Kinetic Energy (KE) at the bottom of the hill. c. How friction and air resistance affect the KE in real situations?	2x3	OR	A mechanic applies 80 N force on a 0.3 m spanner to loosen a tight nut. He increases the distance to 0.6 m by using a longer spanner extension. a. Why the nut becomes easier to loosen with a longer spanner? b. Calculate the amount of force applied on longer spanner to loosen the same nut.	3+3
Q.5	Describe the long-term advantages of using renewable energy sources. Also give the limitations of non-renewable sources. (Three points each)	3+3	OR	Describe any three methods of magnetizing a material.	2x3

— 1SA-I 26004 (B) —

$$\tau = Fd \quad \cdot \quad F = \frac{\Delta p}{\Delta t} \quad \cdot \quad m_1 u_1 + m_2 u_2 = m_1 v_1 + m_2 v_2 \quad \cdot \quad \frac{F_1}{A_1} = \frac{F_2}{A_2} \quad \cdot \quad v = \frac{2\pi r}{T}$$

$$\cdot \quad p = mv \quad \cdot \quad g = 10ms^{-2} \quad \cdot \quad v_f = v_i + at \quad \cdot \quad 2aS = v_f^2 - v_i^2 \quad \cdot \quad P = \rho gh$$



PHYSICS SSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION – A (Marks 12)

Time allowed: 15 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Version No.				
1	2	0	4	1

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4	4	4	●	4
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8	8	8	8	8
9	9	9	9	9

ROLL NUMBER						

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8	8	8	8	8	8	8
9	9	9	9	9	9	9

Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. Which of the following is a derived quantity?	Mass	Speed	Length	Temperature	☐	☐	☐	☐
2. The SI unit of energy is:	Joule	Pascal	Watt	Newton	☐	☐	☐	☐
3. The movement of a pendulum is an example of:	Linear motion	Rotatory motion	Vibratory motion	Circular motion	☐	☐	☐	☐
4. According to Newton's 3 rd law, when someone pushes a concrete wall, the wall:	Does nothing	Moves backward	Pushes back	Pulls forward	☐	☐	☐	☐
5. The branch of physics that studies atomic nuclei and their reactions is known as:	Nuclear Physics	Quantum Mechanics	Thermal Physics	Biophysics	☐	☐	☐	☐
6. A spring extends 3 cm when a 6 N force is applied. What is its spring constant?	18 N/m	200 N/m	50 N/m	2 N/m	☐	☐	☐	☐
7. Which statement best describes a magnetic field?	It is visible in air	It acts only at long distances	It only exists inside metals	It surrounds a magnet	☐	☐	☐	☐
8. A hypothesis is best described as:	A proven law	A testable prediction	A mathematical rule	A universal fact	☐	☐	☐	☐
9. Why does a heavy object resist change in its motion more than a light object?	It has greater inertia	It moves faster	It has more friction	It has more energy	☐	☐	☐	☐
10. A body of mass 4 kg has a kinetic energy of 200 J. What is its speed?	5 m/s	10 m/s	20 m/s	40 m/s	☐	☐	☐	☐
11. Which formula correctly relates force and momentum?	$F = mvt$	$F = \frac{m}{vt}$	$F = \frac{mt}{v}$	$F = \frac{mv}{t}$	☐	☐	☐	☐
12. A temporary magnet is usually made of:	Rubber	Copper	Soft iron	Plastic	☐	☐	☐	☐

—1SA-I 26004-(D)—

• $F = kn$ • $KE = \frac{1}{2}mv^2$ • $p = mv$

**PHYSICS SSC-I**

Inclusive Scheme of Studies 2024 (1200 Marks)

Time allowed: 2:45 Hours**Total Marks Sections B and C: 48****SECTION – B (Marks 30)****Q.2 Attempt the following parts.****(10 x 3 =30)****(Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)**

(i)	Briefly describe the following branches of physics: a. Astronomy b. Particle physics c. Geophysics	1x3	OR	A car travels different distances in different time intervals as shown: Determine. a. Total distance b. Total time c. Average speed	1x3
(ii)	How does an electromagnet improve the working of the electric bell? Describe briefly.	03	OR	How a rapid rise or fall in the liquid column of a barometer help to predict weather changes? Explain briefly.	03
(iii)	A digital balance gives slightly different readings each time same stone is placed on it: a. Identify the error b. How can this error be fixed?	1+2	OR	How the rise of liquid in a drinking straw or dropper justifies that the atmosphere exerts pressure? Explain briefly.	03
(iv)	What is efficiency? Write the formula for percentage efficiency.	2+1	OR	What is meant by velocity? Justify that speed is a scalar quantity while velocity is a vector quantity.	1+2
(v)	Name any three fundamental forces of nature. Briefly describe each force in one sentence.	1x3	OR	Differentiate between precision and accuracy.	03
(vi)	What is average orbital speed? Why increasing the orbital radius increases the total distance traveled in one orbit?	1+2	OR	What is meant by the internal energy of an object? Why heating always increases the internal energy of a substance?	1+2
(vii)	Why areas with high pressure generally have clear skies, while areas with low pressure often experience cloudy or rainy weather? Explain briefly.	1.5 +1.5	OR	State Newton's Third law of motion. Give at least three daily life examples.	1.5 +1.5
(viii)	Compare paramagnetic and dia magnetic materials.	1x3	OR	What is meant by base and derived quantities? Give one example for each.	1.5 +1.5
(ix)	A 2 kg object changes its velocity from 3 ms ⁻¹ to 7ms ⁻¹ in a short time. Calculate the impulse applied on the object.	03	OR	Why soft magnetic materials are preferred for magnetic shielding in laboratories and electrical equipment? Explain briefly.	03
(x)	Why is mercury preferred as thermometric material? Give three reasons.	1x3	OR	What is pressure? Why a sharp knife cuts better than a blunt knife? Explain briefly.	1+2

SECTION – C (Marks 18)**Note: Attempt the following questions.****(3x6=18)****(Use of graph paper is not necessary. Candidates can make their own grid on answer book if required)**

Q.3	A car starts from rest, moves with constant speed, then accelerates, and finally comes to rest. a. Draw distance–time graph. Showing all four stages. b. How the gradient of each section represents the car's speed?	4+2	OR	A player kicks a football of mass 450 g vertically upward at a speed of 22 ms ⁻¹ . a. Neglecting air resistance. How high it would reach? b. If due to air friction ball rises to only 20 m, what is the work done against resistance?	3+3
Q.4	a. What is friction? b. How friction in machine parts leads to energy dissipation as heat? c. Explain two methods to reduce friction.	2+2 +2	OR	Discuss in detail any two applications of permanent magnets.	3+3
Q.5	What is centripetal force? Give its mathematical form. Determine the centripetal force on a 10 kg ball moving in 1.5 m radius circle with a speed of 25 ms ⁻¹ .	1+1 +4	OR	State Pascal's law. How it applies to hydraulic car lift? Explain and show mathematically. Make a sketch.	2+2 +1 +1

— 1SA-I 26004 (D) —

$$\begin{aligned}
 & \bullet \quad K.E = \frac{1}{2}mv^2 & \bullet \quad F = \frac{\Delta p}{\Delta t} & \bullet \quad \frac{F_1}{A_1} = \frac{F_2}{A_2} & \bullet \quad m_1u_1 + m_2u_2 = m_1v_1 + m_2v_2 & \bullet \quad P = \rho gh \\
 & \bullet \quad p = mv & \bullet \quad g = 10ms^{-2} & & & \\
 & \bullet \quad v_f = v_i + at & \bullet \quad 2aS = v_f^2 - v_i^2 & & &
 \end{aligned}$$



PHYSICS SSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

SECTION – A (Marks 12)

Time allowed: 15 Minutes

Candidate Sign. _____

Invigilator Sign. _____

Answer Sheet No. _____

Version No.				
1	9	0	4	1

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8	8	8	8	8
9	●	9	9	9

ROLL NUMBER						

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5	5	5	5	5	5	5
6	6	6	6	6	6	6
7	7	7	7	7	7	7
8	8	8	8	8	8	8
9	9	9	9	9	9	9

Deleting/overwriting is not allowed.
Do not use lead pencil.

Fill the relevant bubble against each question according to curriculum:

Question	A	B	C	D	A	B	C	D
1. Which pair correctly shows base quantity and its unit?	Temperature – kg	Time – m	Mass – kg	Speed – s	☐	☐	☐	☐
2. When a piece of soft iron is placed near a magnet, it becomes:	Permanently magnetized	Temporarily magnetized	Demagnetized	Electrically charged	☐	☐	☐	☐
3. The study of electricity and magnetism together is known as:	Electromagnetism	Nuclear Physics	Classical Mechanics	Particle Physics	☐	☐	☐	☐
4. A car travels 150 m in 10 seconds. What is its speed?	5 m/s	10 m/s	15 m/s	20 m/s	☐	☐	☐	☐
5. A 4 kg object reaches the ground with a kinetic energy of 156.8 J. Ignoring air resistance, from what height was it dropped? (Use $g = 9.8 \text{ m/s}^2$)	2 m	3 m	4 m	5 m	☐	☐	☐	☐
6. Which statement best explains why displacement is a vector quantity?	It has only magnitude	It depends on path length	It has magnitude and direction	It is measured in meters	☐	☐	☐	☐
7. When a north pole of a magnet is placed in an external magnetic field, it:	Always moves toward east	Experiences a force	Becomes a south pole	Stops producing a field	☐	☐	☐	☐
8. A gun recoils backward when a bullet is fired because:	The gun is heavier	Air pushes the gun	The bullet loses speed	Reaction forces act on gun	☐	☐	☐	☐
9. If a bar magnet is cut in half it will become:	The same magnet	Un-magnetized	A monopole	Magnet of less strength	☐	☐	☐	☐
10. In scientific research, a theory is:	A guess with no evidence	A personal opinion	A rule that never changes	Explanation based on experiments	☐	☐	☐	☐
11. A ball dropped freely from a tower reaches ground in 4s, the speed of impact of ball is:	0 m/s	2.45 m/s	19.6 m/s	39.2 m/s	☐	☐	☐	☐
12. A mosquito flying in different directions shows:	Linear motion	Random motion	Circular motion	Vibratory motion	☐	☐	☐	☐

— 1SA-I 26004-(M) —

$$\bullet \quad v = \frac{d}{t} \quad \bullet \quad PE = mgh \quad \bullet \quad g = 9.8 \text{ m/s}^2 \quad \bullet \quad g = \frac{v_f - v_i}{\Delta t}$$



PHYSICS SSC-I

Inclusive Scheme of Studies 2024 (1200 Marks)

26

Time allowed: 2:45 Hours

Total Marks Sections B and C: 48

SECTION – B (Marks 30)

(10 x 3 =30)

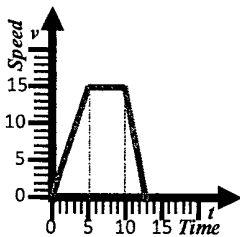
Q. 2 Attempt the following parts.

(i)	Convert the following using appropriate prefixes: a. 0.002 m into mm b. 5×10^{-6} s into ns c. 3000 g into kg	1+1 +1	OR	How much force should be applied on an area of 20 cm^2 to get a pressure 4500 Pa?	03
(ii)	What is meant by precision? How a measurement can be precise but not accurate? (Explain with one example.)	1+2	OR	Discuss the energy conversions when a cyclist goes up to the top of a hill.	03
(iii)	Consider a spring with spring constant 8000 Nm^{-1} . If a force of 500 N is applied to spring. What will be the displacement of spring? Write answer in correct significant figure.	2+1	OR	What is meant by efficiency? Can any machine be 100% efficient? Justify your answer.	1+2
(iv)	How lubrication reduces friction between two surfaces? Explain with the help of an example.	03	OR	Differentiate Ferromagnetic and Paramagnetic materials on basis of domain theory.	1.5 +1.5
(v)	How does the atmospheric pressure vary with altitude? What special measure do the mountain climbers have to take?	1.5+ 1.5	OR	Differentiate qualitatively between rolling and sliding friction.	1.5 +1.5
(vi)	Describe plasma as the fourth state of matter. How the presence of free ions and electrons makes it different from solids, liquids, and gases?	2+1	OR	How different slopes represent three different types of motion in a speed time graph? Make sketches.	1x3
(vii)	A truck of mass 1500 kg moves with a velocity of 10 m/s, while a motorcycle of mass 150 kg moves at 20 m/s. a. Calculate the momentum of each vehicle. b. Which one has greater momentum and why?	1+2	OR	A ball is thrown straight upward with an initial speed of 19.6 m/s. Using $g = 9.8 \text{ ms}^{-2}$: a. After how much time does it reach the highest point? b. What is its velocity at the highest point?	2+1
(viii)	How is Pascal's law used in a hydraulic press to multiply force? Explain briefly.	03	OR	Compare classical mechanics and quantum mechanics. (Any three points)	1x3
(ix)	Differentiate magnetic and non-magnetic materials. (Any three)	1x3	OR	Why internal energy increases when temperature increases? Explain briefly.	03
(x)	Derive the formula for average orbital speed of a satellite.	03	OR	Describe that atmospheric pressure decreases with the increase in height above the earth surface.	03

SECTION – C (Marks 18)

(3x6=18)

Note: Attempt the following questions.

Q.3	Explain the domain theory of magnetism. Discuss the magnetic field of a bar magnet. Also draw its magnetic field lines showing direction.	3+2 +1	OR	a. Describe power as the rate of doing work also write its formula. b. State the SI units of power and work. c. A person does 300 J of work in 10 s. Calculate the power.	2+2 +2
Q.4	a. State newton's Second Law. b. A cart of 15 kg is moving with 25 ms^{-1} . 75 N force is applied to stop the cart. How long does the cart take to stop?	2+4	OR	Explain the principle on which a manometer works. Also describe the construction and working of manometer.	2+2 +2
Q.5	A speed–time graph of a car shows the speed increasing from 0 to 15 m/s in 5 seconds, then staying constant for the next 5 seconds, and finally decreasing back to 0 m/s in 3 seconds. 	2+2 +2	OR	a. What is equilibrium? b. State the difference between stable and unstable equilibrium. c. How the center of gravity affects the type of equilibrium? d. How lowering the center of gravity makes an object more stable? Give example.	1+2 +1 +2
	a. Identify the type of motion in each interval (0–5 s, 5–10 s, 10–13 s). b. Calculate the acceleration during the first 5 seconds. c. Calculate the deceleration during the last 3 seconds.				

— 1SA-I 26004 (M) —

$$F = Kx$$

$$P = \frac{F}{A}$$

$$p = mv$$

$$F = ma$$

$$v_f = v_i + at$$

$$P = \frac{W}{t}$$